
Unit 1 Understanding an Organisation

Structure:

- 1.1 Introduction
 Objectives
- 1.2 Case Study
- 1.3 Definition and Concept of an Organisation
- 1.4 Organisational Process – General
- 1.5 Organisational Process – Vision and Mission
- 1.6 Organisational Process – Strategy
- 1.7 Organisational Process – Structure, System, Process, Jobs, and Tasks
- 1.8 Summary
- 1.9 Glossary
- 1.10 Terminal Questions
- 1.11 Answers

1.1 Introduction

Most of us would have been a part of an organisation and if not a business organisation, at least that of a school. We would also have observed at least small business organisations such as a shop in the neighbourhood, a small factory, etc. Why and how do they work is perhaps something that has never crossed our mind and we take the organisation for granted. We must also have seen some organisations or business thrive and become larger and larger, some remain the same for a long time, and some die. When we become professional managers, we have to have a more intimate knowledge of this so that we can ensure or help in making the organisation in which we work more effective and enable it to grow.

In this unit, we will learn about the organisations with a view to apply this knowledge in making it more effective and prosperous. Our discussion, of course, has a business organisation focus.

1.2 Case Study**Raju's Furniture Business**

Mr. Raju Raman had a family consisting of his wife and two children. He was a carpenter who used to work on daily wages. In the spare time, he used to make tables and sell it to his neighbours and in the local area. After 10 years of hard work as a daily wage labourer who worked for others, he thought he will stop doing it and make and sell tables on a full-time basis. He hired a small room for Rs. 3000 a month, bought some electric saws and other tools with the saving he had made, asked his wife and son to help him. He then went around various furniture shops and told them that he makes tables to order and gave them his mobile number.

Soon orders started to flow and Raju decided to call his two carpenter friends, Shyam and Sanju, to come and work for him on a daily basis but offered them Rs. 350 per day instead of the market rate of Rs. 400. They were happy as they had work daily and could do it in an organised manner under one person. As the work grew, he told them that he will pay them a daily wage of Rs. 350 and an additional of Rs. 50 per table they made. They worked hard and often late into the evening and on an average made Rs. 500 a day as it was easy making a table to clear specifications and using the electric tools rather than sweat it out in the backyard of someone as a casual labourer sawing away with their handsaws. Despite their long hours, they could not complete the order.

Hence, Raju hired 4 more carpenters and made his friend Shyam to take charge of making tables with two newly hired carpenters and Sanju to take charge of making chairs with the other two newly hired carpenters. In addition, he hired Pankaj to buy wood and other fitments, take the finished tables and chairs to various stores, collect the money, pay electricity bills, and other odd jobs outside the shop. The cost of wood suddenly rose, and his traditional timber mill owner explained to him how the increase in cost of diesel has resulted in increased cost of transportation. He could understand that since the tempo owner who takes his finished products to various stores too had hiked the cost of trips. Fortunately, one engineering college was coming up near his shop

and their demand for tables and chairs seemed to be ever increasing. Raju pitched in showing how he made the very same furniture for big furniture shops in the city and how he could provide them these at a lower cost if they gave him bulk orders.

As he saw growth, he put together a team to make new products for the colleges so that they can be modular, strong to withstand the rough use by students, low cost, and with good aesthetics and called this team 'creative team'. Of course, the members of the creative team were select carpenters and his favorite Pankaj who by now had three assistants and had mastered the business of marketing and pricing of the products and book keeping to an extent.

Objectives:

- To analyse the underlying purpose of an organisation.
- To define what an organisation with intrinsic understanding of it.
- To explain the organisational process.

Learning Outcomes

After studying this unit, you should be able to:

- define an organisation.
- identify various components of an organisation.
- evaluate the vision and mission of an organisation.
- explain different types of strategy.
- explain what strategy, structure, systems, processes, jobs and tasks are.

1.3 Definition and Concept of an Organisation

We shall now define what an organisation is. Formally defined, an organisation is a **social system** of people who are structured and managed to meet some **goals**. Organisations are **ongoing** and the structure determines the **relationship** between the functions and positions. Structure also **subdivides roles**, responsibilities, and authority to carry out the tasks. Organisations are **open systems** which are affected by the environment outside its boundary.

Let us now generalise the key words in the definition (which are given in bold) and the business activities for better understanding of the definition.

Social system

The social system was created by Raju and his friends who later became his employees as well as the employees who joined later. In larger businesses, there will be people from diverse backgrounds of education, gender, religion, region, etc. They form the social system. Business is actually something that society has created for itself to make things simpler to fulfill its needs and wants. Therefore, we must accept that it is a social system and it is the social system and not business itself that dictates how a business is to be done at a macro level. This is done through social pressures such as the ones created by NGOs and environmental groups and also through legislation by the representatives of the people, etc. So, though a business is an economic entity, we should not forget that it is the society which gives it sanctity.

Society, through its government, can change the rules of business and say that you can bring in foreign direct investment in this business, but not that business, etc. because a business is an organisation that the society has created to fulfill its needs. If it does not do so, the society tends to control it and even close it down. This is not to say that society and government interferes in everything. In fact, society gives business a lot of freedom, because in doing so, competition takes place, the cost goes down, quality goes up, and society's need is better fulfilled. If you think of mobile phones, you will see how this freedom has helped the society. You can also see how when this led to corruption, another system called the judicial system (also created by the society) intervened on behalf of the society and cancelled about 122 licenses. Every business and young leader of such business which you are or soon will be must understand and accept this power of the society and that business organisations are its creation and will exist only if it fulfills its purpose.

Goal

The goal of the social system in the case we discussed was to make tables and chairs and earn better wages and have better working conditions than working as daily wage carpenters. All businesses, whether big or small, have a goal. Society too has a goal. The goals of society are not only to have tables and chairs but also employment and prosperity. All organisations will have a goal. It may be implicit in small organisations and

often well expressed in larger ones. This is usually done through a vision and mission statement.

Structure and management

There were teams of carpenters and one person for purchase and delivery. Whenever the organisation is large, we need to create a way of doing work systematically. For example, we can have each carpenter buy the wood and materials, make a table and then go and sell it. We can also have one carpenter cut all the pieces of wood to size, another assemble it, a non-carpenter do the purchase and sales, etc. If we group similar work, it is possible for us to be more efficient. So creating a structure is about grouping similar works and doing it. If you recall, Raju as a person, who bought wood, made tables, and sold to the neighbourhood or even the shops in the city, is very different from the Raju who created teams to make tables and chairs and teams to sell them. You can see how structuring has increased the productivity but also brought in other jobs such as specialised jobs of purchasing wood, selling, book keeping, etc.

Ongoing

The business was growing and is intended to go on forever. In fact, though businesses die, they are designed to be ongoing forever. It dies only because we do not care for some of its structures and systems. Had Raju not made the systems and grew, perhaps as he got old, his shop would have closed down too. It is impossible to imagine that his son would have continued to be a one man carpenter after his father, but it is easily possible to imagine his son, running a multi location furniture shop building upon what Raju has created. This is called the 'Ongoing' concept.

Relationship

Relationship existed between the table team head and the two carpenters and the chair team and its head and Raju. You can see that, in larger organisations, there will be several such relationships. Here we are referring to the workplace relationship.

Subdivides the roles

The roles of table making, chair making, purchase, etc. are clear cases of subdividing the roles.

Open System

The cost was negatively affected by fuel price increase and the sale was positively affected by the engineering college, both of which are outside Raju's furniture manufacturing system. This is because it is an open system and is affected by other external systems. This also creates opportunities (engineering college) and threats (increase in the cost of wood and delivery cost of finished goods).

If we have understood what a simple organisation is, let us think of an organisation like Tata. It has over 150 businesses and each business has hundreds of roles and functions and therefore, it is highly complex, but this complex organisation too has the same ingredients as the ones mentioned above. The increased complexity makes it more difficult to manage. Managers and leaders are trained to manage this complexity, but the basic principles remain the same.

Self Assessment Questions

1. An _____ is a social system of people who are structured and managed to meet some goals.
2. The _____ determines the relationship between the functions and positions.
3. Business organisations are open systems which are affected by _____ systems.
4. A _____ is an organisation that the society has created to fulfill its needs.
5. Creating structure is about grouping _____ works and doing it.

1.4 Organisational Process – General

Now that we have learnt the definition of an organisation, let us see the organisational process.

A process is a set of actions that are logically sequenced. As we saw earlier, organisations have goals, and they achieve it by creating products and services, and this is done in a logical sequence. Figure 1.1 depicts the components of this sequence. Later these are explained in some detail. The components of this sequence are:

- Vision and mission
- Strategy

- Structure
- Systems
- Processes
- Jobs
- Tasks (activities)

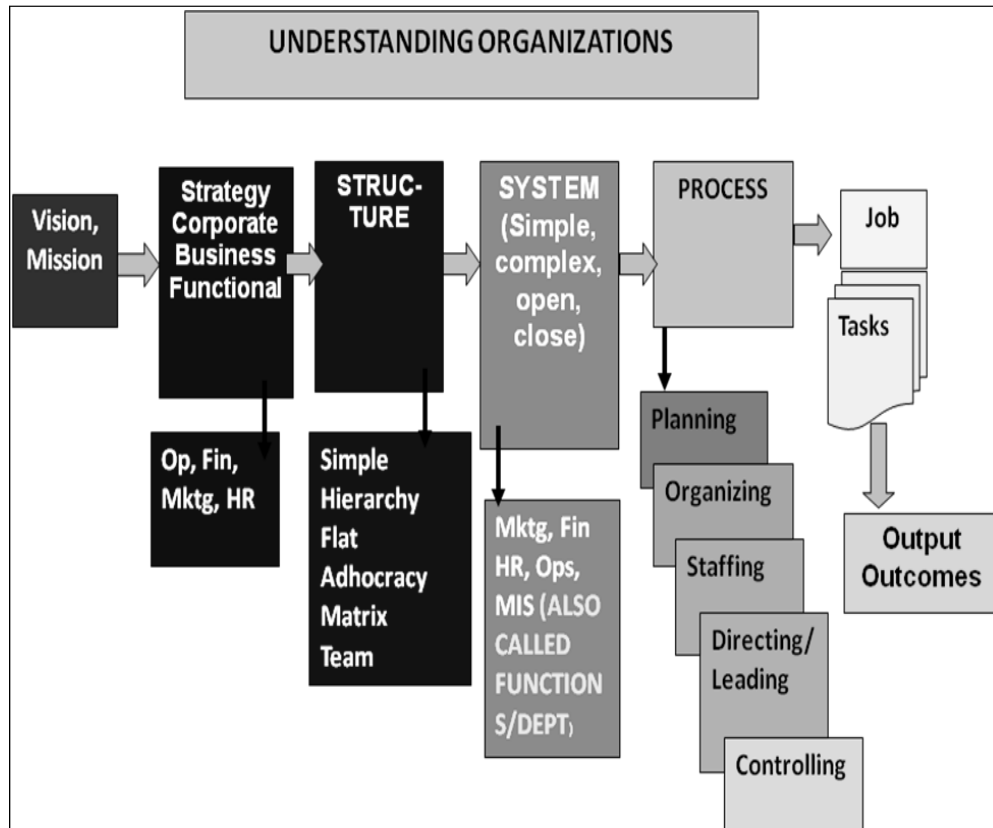


Fig. 1.1: Overview of the organizational Process

(With copyright to reproduce from James P.S)

1.5 Organisational Process – Vision and Mission

Vision and mission

Every organisation has a vision and mission. In the case of Raju, he may not have expressed it in a statement, i.e., it may not be explicit; but perhaps if you had talked with him, you would come to know that he wants to provide faultless tables to the people, earn money, provide for his family, get social

status in the process, etc. A vision statement is a formal statement of what a business wants to be. But who decides what it wants to be? Well, it is not only Raju, but also the customers, the employees, and the society or in other words those who have an interest in the business because they get some benefit out of it. They are called 'stakeholders'. In our case, Raju (owner or promoter or shareholder), the other carpenters (employees), the saw mill (supplier), the shops (distributors), banks who may have given loan (the financiers), and the people who bought the tables and chairs (the customers) are all stakeholders. So the vision cannot be made by the owner alone. He has to fulfill the wishes of the stakeholders. An owner who puts in money has to ask other people before deciding his/her vision. Yes, strange but true.

How do we make a vision?

Well, there has been a lot of research on this and various views. But Collins and Porras in their noted article 'Organisational Vision and Visionary Organisation' spelt out that a vision statement has to have four parts namely:

- 1) Core values,
- 2) Core purposes,
- 3) The Big Hairy Audacious Goal or BHAG (pronounced as 'Beehag') and the vivid description.

Let us now discuss each of these parts in detail.

- The core values are those things very close to your heart that you will not give up at any cost. It can be integrity (for example, I will never cheat on the taxes I have to pay) or quality (I will never use a lower quality wood), etc. Usually we say that you should have only 4 to 6 core values. Of course, personal values and business values may differ. For example, love may be a very important personal value but it may not be so relevant in your business of furniture though it can be relevant if you were having a home for the aged. If you have too many of them, they fail to remain 'core' and loses their impact. Why are core values important? When you have a decision dilemma, they come up like a lighthouse to navigate you.
- Core purpose is the purpose of the organisation, for example, to make furniture. This is something that you want to achieve within the framework of our core values. It gives the achievement orientation to the

business and therefore the focus. When we get an opportunity to expand or sell off and if we are in a decision dilemma, this acts as another light house.

- The BHAG is about having a goal which qualifies the purpose. For example, to be the best furniture maker in the town in five years or to be the most well known furniture supplier in India by 2025, etc. We say that BHAG should be big, yet specific and that your chance of attaining it is only 70%. If the probability of attaining it is more than 70%, perhaps your BHAG is not big enough and not audacious enough and you have set your goals too low. *(Don't read too much into the word 'hairy'. This is an American expression that depicts a huge bear with lot of hair, an animal which is the depiction of power. Children's stories of bear and the toy 'teddy bear' is an American creation perhaps now popular in India too).*
- The vivid description should make the entire vision statement very inspiring to all stakeholders. It should be simple and easily understandable. For example, 'Quit India' is a very simple statement, but it is very vivid and easily understandable by all. Let us be very clear that the vision statement is meant for the stakeholders and therefore, the importance of the vivid and simple statement. We say that your grandmother should be able to understand it or a twelve-year-old child should be able to understand it and you should test it on them. This is also called 'The Grandma's Test' or the '12-Year-Old Test'.

You might ask 'What about the mission statement?' Well, in the Collins and Porras format, the vision and mission are fairly well integrated. Vision is the state that one wants to be in and mission is the way of doing it. You can, of course, say that your vision is to be the No. 1 in India, and you will do it by providing high quality furniture, by maintaining integrity, etc. The Mission statement can be a separate one or it can be rolled into one but Collins and Porras model has been gaining popularity of late because it encapsulates the vision and mission into one and anchors the value.

Activity 1:

Make a vision statement for Raju's furniture business (mentioned in the case). Test the statement as per Grandma's Test.

Hint: Refer Section 1.5

Self Assessment Questions

6. A process is a logically sequenced set of actions. (True/False)
7. Only large and complex organisations have a vision/mission. (True/False)
8. A Big Hairy Audacious Goal should be 100% achievable. (True/False)
9. Vision statements must be elaborate and complex. (True/False)

1.6 Organisational Process – Strategy**Strategy**

Once you have your vision, you have to see how that vision can be executed. Here is where strategy comes in. Strategy means the grand plan and the term is borrowed from the military which distinguished between the grand plan and actual action by calling the former strategy and the latter tactics.

Strategy in organisations can be divided into three:

- Corporate strategy
- Business strategy
- Functional strategy

Let us now discuss each of them in detail.

Corporate strategy

If you are a company like TATA or BIRLA, you will have several businesses under you and a corporate headquarters which controls these. Each of these businesses may be run by an independent company much like Tata Motors runs the vehicle business and Tata Consultancy Service runs the IT business (they both are different companies). The corporate headquarters will have grand plans on how each business should operate. For example, it might say that that Tata Motors should design, develop, and manufacture an indigenous small car Nano or that TCS should expand business to China.

This grand plan created by the corporate headquarters is called 'Corporate Strategy'. It lays down the grand plan to be followed by each business under a corporate headquarters. If you are small like Raju, you will not have a corporate strategy; however, will still have a strategy for your business. Though you can call it the corporate strategy of Raju on the argument that

Raju is a corporate, it is perhaps inappropriate. Since Raju is not very big, usually we prefer to call his strategy as business strategy.

Business strategy

Every business, small or big, will have a business strategy. This is the grand plan for doing business. For example, Raju can make a grand plan of expanding his business to another district or having branches. He can take a decision to be the provider of low cost furniture or highly differentiated furniture, household furniture only, modular furniture only, a supplier of all furniture needs, etc. This is what business strategy is. If you were a TATA company, each business of TATA (say Tata Motors, Tata Iron and Steel Company, Tata Hotels, etc) would have their own business strategy in consonance with the corporate strategy.

Functional strategy

Once you have a business strategy, each independent department will need its own strategy. For example, the HR of Tata Motors will need a strategy quite different from the HR of Tata Consultancy Service (TCS). So, the operations, marketing, finance, HR, etc of each business will make their own grand plan or strategy and this is called functional strategy. In the case of Raju, he might plan to hire more carpenters and pay daily wages or wages per piece. This is the HR strategy (of course it is only one element of HR strategy but illustrates what functional strategy means). In other words, the grand plan made by each functional area is called functional strategy. Usually organisations have operations strategy, finance strategy, marketing strategy, and HR strategy. If you are large and well diversified, you may have an information strategy, supply chain strategy, etc. also and this depends to some extent on the nature of your business. For example, if you are in retail chain business like Big Bazaar, you may have a merchandise strategy (what products to sell) and supply chain strategy (how to procure the products at the lowest cost and move them nationally) as these are very important to business success. Similarly, Raju too can have a strategy to purchase wood in bulk from Malaysia and ship it and call it his procurement strategy.

Self Assessment Questions

10. Strategy can be broadly divided into _____ strategy, _____ strategy and _____ strategy.

11. _____ lays down the grand plan to be followed by each business under a corporate headquarters.
12. When each independent department creates its own strategy, it is known as_____.

1.7 Organisational Process – Structure, System, Process, Jobs, and Tasks

Structure

Structure follows strategy. In the case of Raju, we found that Shyam and his team takes care of the table line of production, Sanju and his team the chair line, and Pankaj, the administrative activities. These being small worked directly under Raju. However, if this was a large operation (let us assume that Raju will grow real big one day), he probably would need HR people to recruit, select, plan compensation, manage pay rolls and legal issues, finance people to manage the financial aspects, etc. Thus the production, HR, finance and some people to do marketing, etc. becomes the pillars of his business.

You would have seen these basic pillars in most modern businesses. Each of these will have a head, some people, a rule for communication, i.e., whether you have to say things only to your immediate boss or you can (and should) do that to several people involved in the action at the same time, etc. These basic pillars on which the business is built are called the structure.

Structures can be:

- 1) simple (just like the present case of Raju),
- 2) hierarchical (when the rule says that you can only communicate to your immediate boss or subordinate as in the case of a government department
- 3) Flat (when you can communicate to your boss and laterally to others on a need basis),
- 4) adhocracy (the opposite of bureaucracy). It is considered an organisation structure of the future where information flow takes place in all directions thus enabling an organisation to take advantage of opportunities as never before),

- 5) Matrix (when you belong to a basic pillar, for example, finance and report to project manager for the project finance component of it and global product manager as part of product development team, and
- 6) Team (Which is a unique and strongly bonded entity that comes together for a function such as product development team such as the 'Creative Team' of Raju).

Systems

We usually hear the complaint 'there is no system here and it is all a one man show' or at the other end of the spectrum, the complaint that 'everything here is system driven'.

So, what is this system? The easiest way to think of a system is the way we group functions, create a logical order, organise the flow of information and resources and let it run almost by itself. Each system is somewhat independent such as an HR system or marketing system but each system is also highly interdependent on other system, for example, the marketing system is dependent on HR system to procure good marketing people and on the operation system to ensure that it gets high quality products to sell. It is easier to think of human body system consisting of the cardiovascular system, digestive system, etc and make an analogy. Cardiovascular system is fairly independent but it is also dependant on other systems for energy, for example, the respiratory system for oxygen.

In business too, we have systems such as HR, marketing, operations, finance, etc. and can have more systems such as logistics, supply chain, international marketing, etc. depending on the need. The role of a system is to take in some input and create some output. The creating part of it is called throughput.

Thus, typically every system would have:

- 1) input (raw materials, for example),
- 2) throughput (the way it is mixed or connected), and
- 3) output (a final or interim product which becomes raw material for another input).

Steel, for example, uses coal and iron ore, the high heat and mixing is the throughput and the final output is steel which in itself is of no use to anyone but becomes a raw material to make cars, railway tracks, or build houses.

The way systems work is studied in detail and systems theory is a well known theory. We will not go deep into system theory but by now we would have understood the basics of a system.

Processes

Process is a set of logical activities that lead to some final or interim output. For example, taking pieces of wood, making it smooth, cutting it, making grooves, connecting them, finishing them, and polishing them are processes to create a table. These have to be done in some logical sequence. This is what a process is. Let us now look at Raju's business itself. He has to manage things and for this, he will have to plan the number of pieces to be made, how much wood to buy, which furniture to make, how to organise these, where to get manpower, etc. These too have to be done in some logical sequence. For example, he has to plan and decide the number of tables before deciding on the quantity of wood. Thus at the business level, some processes come into play. These are planning, organising, staffing, leading, and controlling. We call them management processes. In other words, in the management system, these processes convert the input into output (furniture, profit, growth, etc.). You can intuitively see that if you consider manufacturing table as a system, then the throughput is created through the processes of cutting wood, etc. while if you consider the business as a system, planning, organising, etc becomes the processes.

Jobs

Each process is made of several jobs. A job is a set of similar activities. For example, the job of a teacher involves the task of preparing for the class, teaching, making question papers, correcting papers, giving feedback, etc. Of course, you can see that some of it can be done by another teacher. For example, setting question papers or correcting papers. So the job is a cluster of similar tasks. The size of this cluster depends on the capacity of a person to do it. If you are a small business school, a teacher may do all those jobs but if it is a university with some 20,000 students studying one program such as MBA, then many people will do one task.

Tasks (activities)

This is perhaps the most basic unit in an organisation. In the example given above, teaching can be a task, cutting wood can be a task, and making grooves in the wood can be a task.

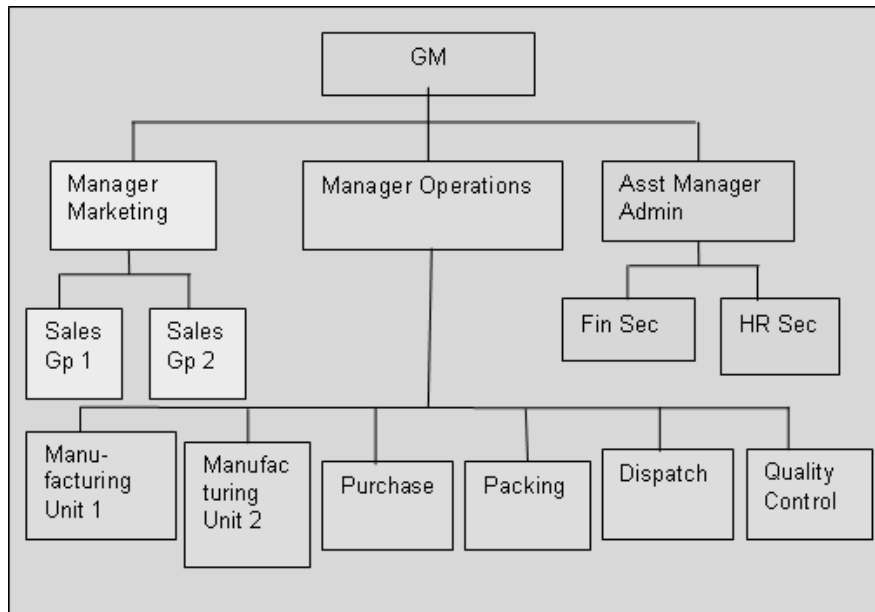
Thus, we have learnt that every organisation has a vision and mission. To execute the organizational vision we create strategies. Structures, systems and processes help the organisation execute the strategies.

Activity 2:

Suppose you are working in a small industry which manufactures stabilisers. You are well known for your quality and have a quality assurance department of 5 people and a manager and a sales team of one senior manager and 18 sales staff. The HR department is rather small with 3 staff and no manager and also the finance with only 4 people. They report to the General Manager directly and the GM takes care of the manufacturing as well. There are 60 people in the manufacturing which includes packing, purchase, and dispatch in addition to the actual manufacturing process.

What sort of structure are you likely to have? Justify your answer.

Hint: Refer Section 1.7. You may also refer the below structure.

**Self Assessment Questions**

13. The basic pillars on which the business is built is called_____.

14. _____ is a set of logical activities that lead to some final or interim output.
15. _____ is a basic unit or activity in an organisation.

1.8 Summary

Let us recapitulate the important concepts discussed in this unit:

- Organisation is a social system of people who are structured and managed to meet some goals.
- Every organisation should have a vision and mission or in simpler language a purpose.
- To achieve the vision, a lot of activities or tasks such as the task of buying wood or cutting it have to be performed.
- Similar tasks, when grouped together, become the job.
- When jobs are grouped together and put into some logical sequence and it becomes the process.
- Processes use input and convert these into an output.
- The input, the logical processes along with the output is called a system. For example, the manufacturing system of tables and chairs has an input of wood, nails, knowledge of the carpenter, etc, a series of processes or throughput such as cutting the wood, making the grooves, connecting them, and the output. The input of wood comes only because someone has purchased it and brought it to the location and this is based on the purchase system. So, the output of the purchase system is the input for the manufacturing system and that of the manufacturing for the marketing system, etc.
- It follows that there has to be some internal communication and control within the purchasing system and between the purchasing system and manufacturing system, etc. This is called structure. Structure controls the way we interact within and in between systems.
- The structure depends on the strategy, i.e., if the strategy is to make the chair making very innovative, then everyone need to talk to everyone else and act accordingly but if it is to be made very cost effective and run like a machine without much innovation, then this type of communication is not necessary and you may adopt a hierarchical system. The strategy itself is based on the vision and mission.

Why This Reiteration?

We have reiterated this because of its overwhelming importance to understand business as such.

- When we create change, we actually make a change to one of the above elements such as strategy, structure, etc. If we change one of them, what follows in the hierarchy will also undergo a change usually, i.e., if we change the strategy, usually it will be necessary to make adjustments in the structure, systems, processes, etc.
- If you understand this serial effect, you will also appreciate that when we try to solve an organisational problem, you will be able to do it at the appropriate level, i.e., if you have a system problem, solve it at that level and don't try to fix the structure. If you solve at a higher level than required, then you will be wasting resources as you will have to adjust the lower levels too and if you solve at the lower level, the problem will persist as the issue will not be completely addressed.
- In a fast changing world, the leaders should understand this in order to pitch change at the appropriate level.

1.9 Glossary

- **Process:** The series of logical actions in a system that converts the input into an output.
- **Strategy:** The grand plan which may pertain to the corporate, business or functional area such as marketing. This is a way of achieving the vision and mission.
- **Structure:** The way various departments are organised. It indicates how these departments communicate with each other.
- **System:** It indicates how the input is converted into output using a series of processes together called the throughput. Systems are independent yet interdependent.
- **Vision and mission:** A formal definition of the purpose of the organisation and what it wants to achieve. Usually it is in a written form and becomes a lighthouse for the organisation to navigate itself.

1.10 Terminal Questions

1. Briefly justify business as a social system.
2. Describe the concept of vision and mission in an organisation.
3. What do you understand by strategy? Briefly explain functional strategy.
4. Explain the different types of structures found in organisation.

1.11 Answers**Self Assessment Questions**

1. Organisation
2. Structure
3. External
4. Business
5. Similar
6. True
7. False
8. False
9. False
10. Corporate, functional, business
11. Corporate
12. Functional
13. Structure
14. Process
15. Tasks

Terminal Questions

1. Business is something that society has created for itself to make things simpler to fulfill its needs and wants. The society exercises control over business by changing rules of business Refer Section 1.3 for more details.
2. Vision is the state that an organisation wants to be in and mission is the way of doing it. Refer Section 1.3 for more details.
3. Strategy is the broad plan through which the organisation achieves its vision. It may be divided into corporate strategy, functional strategy and business strategy. Refer Section 1.6 for more details.

4. The different types of structures found in organisation are flat, simple, hierarchical, adhocracy, matrix, team. Refer Section 1.7 for more details.

References/E-Reference:

- Bower, M., (1979) *Perspective on McKinsey*, New York, NY: McKinsey & Company, Inc, Pg. 234.
- Gibson, P., (1979) *Being Good Isn't Enough Anymore*, Forbes, Pg. 40.
- Harold, K. and Heinz, W., (2010), *Essentials of Management – An International Perspective*, Tata McGraw Hill, New Delhi.
- James, C. C., and Jerry, P. I., (1991), *Organizational Vision and Visionary Organization*, California Management Review, 10 Oct 1991.
- Lacey, R., (1986) *Ford: The Men and the Machine*, New York, NY: Ballantine Books, Pg. 93.